

Dynamic Modelling of the Non-Catalytic Process of Ethylene Oxide Hydrolysis

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Abstract

A non-catalytic process of ethylene oxide (EO) hydrolysis producing monoethylene glycol (MEG), diethylene glycol (DEG), and triethylene glycol (TEG) has been considered. The reactor unit consists of three basic elements: a stirred mixing tank, a heat exchanger and a tubular reactor. Mathematical models of these three basic elements have been developed. Material balances on ethylene oxide and water in the stirred mixing tank have been discussed. Energy balances on liquid and vapor phases in the heat exchanger have been presented, while steady-state equations of the tubular reactor have been developed. In order to investigate an optimal control system of this process an adequate dynamic model of ethylene oxide hydrolysis process has been proposed. Multivariable block-flow diagram of the process dynamics has been proposed. Using a standard procedure of linearization, a linear model of the reactor unit in the transform domain has been obtained.

Keywords: Dynamic Modelling, reactor unit, ethylene oxide hydrolysis.

1. Introduction

One of the most relevant processes of organic synthesis is the production of monoethylene glycol (MEG) and its analog: di-, tri-, tetra-, and polyethylene glycols. In USA the ethylene glycols constitutes about 38% of overall organic synthesis production^[1].

The U.S. production of ethylene oxide (EO) is over 2.5 metric tons, of which nearly 60% is hydrolyzed to ethylene glycol. About half of the produced glycol is used in the production of polyethylene terephthalate, and a high proportion of the remainder is the basis for antifreeze production. Di- and triethylene glycols are used in resins and for gas drying processes^[2].

Ethylene oxide reacts with water to give glycol and/or polyglycols. Several companies have sought direct routes from ethylene to ethylene glycol, to achieve higher selectivity. The non-catalytic process of (EO) hydrolysis is the most widely used method in ethylene glycol synthesis, because it is simple and has an economic process^[3].

The kinetics of this process has been investigated by the same author in a recent work^[4]. The kinetic equations describing the non-catalytic process of ethylene oxide hydrolysis and the consecutive-parallel reactions generally have been previously published^[4-5]:

$$\frac{dX}{d\tau} + K \cdot X [Y + m_1 \cdot X_1 + m_2 \cdot X_2 + m_3 \cdot X_3 + \dots] = 0$$

$$\frac{dY}{d\tau} + K \cdot X \cdot Y = 0$$

$$\frac{dX_1}{d\tau} + K \cdot X \cdot [m_1 \cdot X_1 - Y] = 0$$

$$\frac{dX_2}{d\tau} + K \cdot X \cdot [m_2 \cdot X_2 - m_1 \cdot X_1] = 0$$

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With initial conditions: $X|_{\tau=0} = X^0$, $Y|_{\tau=0} = Y^0$, $X_i|_{\tau=0} = X_i^0$.

Where: X and Y – concentration of initial reactants: water and ethylene oxide respectively.

X_i – concentration of reaction products: mono-, di-, tri-, tetraethylene glycols,

$$m_i = \frac{K_i}{K} = \text{const} \quad \text{i.e.}$$

the rate constants of all reaction steps except the first one are equal ^[5].

The process is carried out in a tubular reactor with ideal plug flow and the structure of the reactor model is defined by the structure of the kinetic reaction model^[6]. Studying regimes of the reactor operation makes it possible to determine concentrations of the input and the output materials on the basis of indirect measurements, which allows to monitor these concentrations, as well as to minimize energy consumption of the process. The solution of this task can be achieved by applying modern control theory^[7].

Although most of existing control theory only applies to linear systems with constant coefficients, the great majority of practical process control problems including this study involve nonlinear system. The easiest and the most common approach for designing the control schemes for nonlinear multivariable systems is to linearize the modeling equations and apply standard linear design procedures^[7, 8].

In this work an adequate dynamic model of ethylene oxide hydrolysis process has been proposed in order to investigate an optimal control system of this process.

2. Process description

Fig.1 shows the simplified schematic flowchart of the industrial process of ethylene glycols synthesis. Ethylene oxide and water are fed continuously to the stirred mixing tank through different distributors. These substances are heated in the heat exchanger up to the operating reactor temperature. After reaching this temperature the reaction mixture were fed to the reactor. The hydrolysis reactor was a tubular adiabatic U shaped reactor of length 150m and diameter of 0.53m. First some distillation columns are intended for evaporation of water. The remaining columns are intended for series separation of reaction products: mono-, di-, tri-, tetraethylene glycols and so on. The evaporated water was recycled to the hydrolysis reactor.

3. Development of dynamic model equations

The first step in this study was to develop a simple yet reliable mathematical model of the reactor unit. As shown in Fig. 1 the reactor unit of the non-catalytic process of ethylene oxide hydrolysis consists of three basic elements: a stirred mixing tank, a heat exchanger and a tubular reactor. Let's consider the mathematical models of these three elements.

3.1 The stirred mixing tank

The total material balance can be written as:

$$\frac{dV}{d\tau} = U_x + U_y + \sum_{i=1}^n U_i - U \quad (1)$$

With initial condition: $V|_{\tau=0} = V^0$

Where: V – volume of the reactor; τ – time of the process; U – volumetric flowrate of fluid on the output of the tank; U_x , U_y , U_i – volumetric flow rate of ethylene oxide, water and ethylene glycols respectively.

If we then define as U_x^0 , U_y^0 , U_i^0 the steady-state values of U_x , U_y , U_i , that is, satisfying

$$0 = U_x^0 + U_y^0 + \sum_{i=1}^n U_i^0 - U^0 \quad (2)$$

Then one can use the deviation variables:

$$\begin{aligned} V &= V^0 + \Delta V & U_x &= U_x^0 + \Delta U_x & U_y &= U_y^0 + \Delta U_y \\ U_i &= U_i^0 + \Delta U_i & U &= U^0 + \Delta U \end{aligned}$$

Equation (1) can be represented as:

$$\frac{d\Delta V}{d\tau} = \Delta U_x + \Delta U_y + \sum_{i=1}^n \Delta U_i - \Delta U, \quad \Delta V|_{\tau=0} = 0 \quad (3)$$

The transfer functions can be obtained from this equation as follows:

$$\frac{\Delta \bar{V}}{\Delta \bar{U}_x} = \frac{1}{s}; \quad \frac{\Delta \bar{V}}{\Delta \bar{U}_y} = \frac{1}{s}; \quad \frac{\Delta \bar{V}}{\Delta \bar{U}_i} = \frac{1}{s}; \quad \frac{\Delta \bar{V}}{\Delta \bar{U}} = -\frac{1}{s} \quad (4)$$

Where: $\Delta \bar{V}$, $\Delta \bar{U}_x$, $\Delta \bar{U}_y$, $\Delta \bar{U}_i$, $\Delta \bar{U}$ are the Laplace transform of ΔV , ΔU_x , ΔU_y , ΔU_i , ΔU , respectively.

The material balance on ethylene oxide can be written as[8]:

$$\frac{d(V \cdot X_{in})}{d\tau} = U_x \cdot \langle X_{in} \rangle - U \cdot X_{in} \quad (5)$$

Where: $\langle X_{in} \rangle$ -concentration of ethylene oxide on the input of the tank; X_{in} - concentration of ethylene oxide on the output of the tank.

If we then define as U_x^0 , $\langle X_{in}^0 \rangle$, U^0 , and X_{in}^0 the steady-state values of U_x , $\langle X_{in} \rangle$, U , and X_{in} that is, satisfying:

$$0 = U_x^0 \cdot \langle X_{in}^0 \rangle - U^0 \cdot X_{in}^0 \quad (6)$$

Clearly the tank model is nonlinear; however, it is possible to linearize the equation about the desired operation point by using Taylor series expansions[9] truncated after the first-order terms. The linearized equation can be presented as:

$$X_{in}^0 \cdot \frac{d\Delta V}{d\tau} + V_0 \cdot \frac{d\Delta X_{in}}{d\tau} = U_x^0 \cdot \Delta \langle X_{in} \rangle + \langle X_{in}^0 \rangle \cdot \Delta U_x - U_0 \cdot \Delta X_{in} - X_{in}^0 \cdot \Delta U \quad (7)$$

This equation describes three dynamic channels with input variables $\Delta\langle X_{in} \rangle$, ΔU_x , ΔU and output variable ΔX_{in} .

Let's consider the first dynamic channel: $\Delta\langle X_{in} \rangle \rightarrow \Delta X_{in}$.

In this case $\Delta V = 0$, $\Delta U_x = 0$ and $\Delta U = 0$. The transfer function can be written as:

$$\frac{\Delta \bar{X}_{in}}{\Delta \langle \bar{X}_{in} \rangle} = \frac{U_x^0 / U^0}{T \cdot s + 1} \quad (8)$$

$$\text{Where: } T = \frac{V^0}{U^0}$$

Let's consider the second dynamic channel: $\Delta U_x \rightarrow \Delta X_{in}$.

In this case $\Delta\langle X_{in} \rangle = 0$ and $\Delta U = 0$. Knowing from equation (4) that, the $\Delta \bar{V} = \frac{\Delta \bar{U}_x}{s}$ transfer function can be presented as:

$$\frac{\Delta \bar{X}_{in}}{\Delta \bar{U}_x} = \frac{(\langle X_{in}^0 \rangle - X_{in}^0) / U^0}{T \cdot s + 1} \quad (9)$$

Let's consider the third dynamic channel: $\Delta U \rightarrow \Delta X_{in}$.

In this case $\Delta\langle X_{in} \rangle = 0$ and $\Delta U_x = 0$. Knowing from equation (4) that $\Delta \bar{V} = -\frac{\Delta \bar{U}}{s}$, the transfer function can be written as:

$$\frac{\Delta \bar{X}_{in}}{\Delta \bar{U}} = 0 \quad (10)$$

This equation is consistent with the physical concrete as the underflow has no influence on the concentration in the tank.

Similarly it is possible to investigate and to describe dynamics of remaining channels of the tank.

The material balance on water can be presented as:

$$\frac{d(V \cdot Y_{in})}{d\tau} = U_y \cdot \langle Y_{in} \rangle - U \cdot Y_{in} \quad (11)$$

Where: $\langle Y_{in} \rangle$ - concentration of water on the input of the tank; Y_{in} - concentration of water on the output of the tank.

If there is recycles it is necessary to consider the material balance on mono-, di-, tri-, tetra- and polyethylene glycols as follows:

$$\frac{d(V \cdot X_{i \text{ in}})}{d\tau} = U_i \cdot \langle X_{i \text{ in}} \rangle - U \cdot X_{in}, \quad i = 1, 2, 3, \dots \quad (12)$$

3.2 The heat exchanger

The heat exchanger represents a shell and a tube heat exchanger. The process fluid flows through the tubes and the vapor flows inside the cylindrical shell. Dynamics of such objects is well enough investigated^[9,10].

Let's consider that liquid flows in tubes in ideal plug-flow fully-developed regime. The energy balance on liquid phase can be written as:

$$\frac{1}{\lambda} \cdot \frac{dT}{d\tau} + \frac{\omega}{\lambda} \cdot \frac{dT}{dz} = T_V - T, \quad \text{where:} \quad \lambda = \frac{A \cdot K}{\sigma \cdot C_p \cdot \rho}$$

Where: σ , C_p , ρ , A , ω , K , T , T_V are tube surface area, isobaric heat capacity of the liquid, liquid density, heat transfer area, linear velocity of the liquid, heat transfer coefficient, temperature of the reaction mixture and vapor temperature respectively.

The energy balance on vapor can be presented as:

$$C \cdot \frac{dT_V}{d\tau} = G \cdot (H_{0V} - H_{0C}) - A \cdot K \cdot l \cdot (T_V - T) - G \cdot T$$

Where: C , G , $(H_{0V} - H_{0C})$, l are equivalent integral heat capacity of the heating agent, volumetric flow rate of the vapor, enthalpy difference between vapor and condensate, and length of the heat exchange zone respectively.

The equations can be aggregated in the following system:

$$\begin{aligned} \frac{dT}{d\tau} + \omega \cdot \frac{dT}{dz} &= \lambda \cdot (T_V - T) \\ T_V^0 \cdot \frac{dT_V}{d\tau} + T_V &= \theta \cdot G + T \end{aligned} \quad (13)$$

$$\text{Where: } \theta = \frac{H_V - H_C}{K \cdot l \cdot A}; \quad T_V^0 = \frac{C}{K \cdot l \cdot A}.$$

$$\text{The boundary conditions: } T|_{\tau=0} = T^0; \quad T|_{z=0} = \langle T_{in} \rangle; \quad T_V|_{\tau=0} = T_V^0$$

The steady-state equation can be written as:

$$\omega^0 \cdot \frac{dT}{dz} = \lambda \cdot \theta \cdot G^0; \quad T^0|_{z=0} = \langle T_{in}^0 \rangle$$

The solution of this equation (13) represents a linear function of z :

$$T^0 = \frac{\lambda \cdot \theta \cdot G^0}{\omega^0} \cdot z + \langle T_{in}^0 \rangle \quad (14)$$

The linearized equations can be presented as:

$$\begin{aligned} \frac{d\Delta T}{d\tau} + \omega^0 \cdot \frac{d\Delta T}{dz} + \Delta\omega \cdot \frac{dT^0}{dz} &= \lambda \cdot (\Delta T_V - \Delta T) \\ T_V^0 \cdot \frac{d\Delta T_V}{d\tau} + \Delta T_V &= \theta \cdot \Delta G + \Delta T \end{aligned} \quad (15)$$

$$\text{With initial conditions: } \Delta T|_{\tau=0} = 0; \quad \Delta T|_{z=0} = \Delta \langle T_{in} \rangle; \quad \Delta T_V|_{\tau=0} = 0$$

By application of the Laplace transform^[11, 12], the system which is described by equation (15) can be represented as follows:

$$\begin{aligned} \omega^0 \cdot a(s) \cdot \frac{d\Delta\bar{T}}{dz} + b(s) \cdot \Delta\bar{T} &= c(s) \cdot \Delta\bar{\omega} + d \cdot \Delta\bar{G} \\ \Delta\bar{T}(s, z)|_{z=0} &= \Delta\langle\bar{T}_{in}\rangle(s) \end{aligned} \quad (16)$$

$$\text{Where: } a(s) = T_V^0 \cdot s + 1; \quad b(s) = [(s + \lambda) \cdot (T_V^0 \cdot s + 1) - \lambda];$$

$$c(s) = -\frac{\lambda \cdot \theta \cdot G^0}{\omega^0} \cdot (T_V^0 \cdot s + 1); \quad d = \lambda \cdot \theta$$

To investigate the dynamic channels: $\Delta\omega \rightarrow \Delta T$ and $\Delta G \rightarrow \Delta T$ it is necessary to accept zero boundary conditions in (16), and to consider $\Delta\bar{\omega}(s, z)$ and $\Delta\bar{G}(s, z)$ as constants in z . Thus, two solutions can be obtained as follows:

$$\Delta\bar{T}(s, z) = \Delta\bar{\omega}(s) \cdot \frac{c(s)}{b(s)} \cdot \left[1 - \exp\left(-\frac{b(s)}{a(s)} \cdot \frac{z}{\omega^0}\right) \right] \quad (17)$$

$$\Delta\bar{T}(s, z) = \Delta\bar{G}(s) \cdot \frac{d}{b(s)} \cdot \left[1 - \exp\left(-\frac{b(s)}{a(s)} \cdot \frac{z}{\omega^0}\right) \right] \quad (18)$$

If the exponential in equations (17, 18) is expanded by the means of a series truncated up to the second term of the series, and letting $z = l$, $\frac{1}{\omega^0} = \tau_D$ and $\Delta\bar{T}(s, l) = \Delta\bar{T}_{in}$, then the transfer functions can be written as:

$$\frac{\Delta\bar{T}_{in}(s)}{\Delta\bar{\omega}(s)} = \frac{c(s)}{a(s)} \cdot \tau_D = -\lambda \cdot \theta \cdot G^0 \cdot \tau_D \quad (19)$$

$$\frac{\Delta\bar{T}_{in}(s)}{\Delta\bar{G}(s)} = \frac{d}{a(s)} \cdot \tau_D = \frac{\lambda \cdot \theta \cdot \tau_D}{T_V^0 \cdot s + 1} \quad (20)$$

To investigate the dynamic channel $\Delta\langle T \rangle_{in} \rightarrow \Delta T$ in equation (16) let's assume $\Delta\omega = 0$ and $\Delta G = 0$.

Solving it at non-zero boundary conditions i.e. $\Delta\bar{T}(s, z)|_{z=0} = \Delta\langle\bar{T}\rangle_{in}(s)$, yields:

$$\frac{\Delta\bar{T}_{in}(s)}{\Delta\langle\bar{T}\rangle_{in}(s)} = \exp\left(-\frac{b(s)}{a(s)} \cdot \tau_D\right) \quad (21)$$

$$\text{Where: } \frac{b(s)}{a(s)} = \frac{T^0 \cdot s^2 + (1 + \lambda \cdot T^0) \cdot s}{T^0 \cdot s + 1}$$

The term conforming pure time-delay is:

$$\left[\frac{b(s)}{a(s)} - s \right] + s = \frac{\lambda \cdot T^0 \cdot s}{T^0 \cdot s + 1} + s$$

Thus, the transfer function can be written as:

$$\frac{\Delta \bar{T}_{in}(s)}{\Delta \langle \bar{T} \rangle_{in}(s)} = \exp\left(-\frac{\lambda \cdot T^0 \cdot s}{T^0 \cdot s + 1} \cdot \tau_D\right) \cdot \exp(-s \cdot \tau_D) \quad (22)$$

The transcendental transfer function, $W(s) = \exp\left(-\frac{\lambda \cdot T^0 \cdot s}{T^0 \cdot s + 1} \tau_D\right)$, is approximated by a periodic unit of the first order in the form $W_A(s) = \frac{K}{T \cdot s + 1}$ using a method of integral estimations^[8], previously having determined a gain K.

$$\lim_{s \rightarrow 0} W(s) = 1$$

$$\lim_{s \rightarrow 0} W_A(s) = K$$

Therefore $K=1$.

Let's consider the first integral estimation:

$$\lim_{s \rightarrow 0} \frac{dW(s)}{ds} = -\lambda \cdot T^0 \cdot \tau_D$$

$$\lim_{s \rightarrow 0} \frac{dW_A(s)}{ds} = -T$$

$$\text{Therefore } T = \lambda \cdot T^0 \cdot \tau_D$$

The simplified transfer function can be represented as:

$$\frac{\Delta \bar{T}_{in}(s)}{\Delta \langle \bar{T} \rangle_{in}(s)} = \frac{1}{\lambda \cdot T^0 \cdot \tau_D \cdot s + 1} \cdot \exp(-s \cdot \tau_D) \quad (23)$$

3.3 The tubular reactor

The reactor model considered in this work consists of two main sections: The steady state equations and the dynamic equations. According to Shargardski^[6] it is possible to consider the reactor as an ideal plug-flow reactor with sufficient degree of accuracy. There is no possibility to create an operative automatic control of the reaction mixture composition, but the temperature can be measured only in automatic mode. Therefore, we shall consider only equations of concentration change of ethylene oxide and ater.

The steady state equations of the tubular reactor can be obtained from the kinetics model by replacing τ with $\frac{z}{\omega_0}$. Where: z – spatial position; ω_0 – linear velocity of liquid. Thus, the boundary conditions of the obtained system of equations can be determined by initial conditions of equation's system of kinetics.

The steady-state equations of the tubular reactor can be presented as follows:

$$\begin{aligned} -\omega^0 \cdot \frac{dX^0}{dz} &= K \cdot X^0 \cdot [Y^0 + m \cdot (Y_{in}^0 - Y^0)] \\ -\omega^0 \cdot \frac{dY^0}{dz} &= K \cdot X^0 \cdot Y^0 \\ -\omega^0 \cdot \frac{dT^0}{dz} &= -\alpha \cdot K \cdot X^0 \cdot [Y^0 + m \cdot (Y_{in}^0 - Y^0)] \end{aligned} \quad (24)$$

Where: $\alpha = \frac{(-\Delta H)}{C_p \cdot \rho}$; index “0” means a constant value; index “in” means value in an input stream.

The object is non-linear on spatial position and to obtain the linear mathematical description a considerable simplifications are necessary. According to the system of equation (24) it is possible to obtain the following:

$$\frac{dT}{d\tau} + \omega^0 \cdot \frac{dT}{dz} = -\alpha \cdot \left[\frac{dX}{d\tau} + \omega^0 \cdot \frac{dX}{dz} \right] \quad (25)$$

The steady state equation can be presented as:

$$\omega^0 \cdot \frac{dT^0}{dz} = -\alpha \cdot \frac{dX^0}{dz} \quad (26)$$

Equation (25) can be represented as:

$$\frac{d\Delta T}{d\tau} + \omega^0 \cdot \frac{d\Delta T}{dz} = -\alpha \cdot \left[\frac{d\Delta X}{d\tau} + \omega^0 \cdot \frac{d\Delta X}{dz} \right] \quad (27)$$

At boundary conditions: $\Delta X(\tau, z)|_{\tau=0} = 0$; $\Delta X(\tau, z)|_{z=0} = \Delta X_{in}(\tau)$;
 $\Delta T(\tau, z)|_{\tau=0} = 0$; $\Delta T(\tau, z)|_{z=0} = \Delta T_{in}(\tau)$.

By application of the Laplace transform^[11] along two axes, equation(27) can be written as:

$$(\omega^0 \cdot s + p) \cdot \widehat{\Delta T}(s, p) - \omega^0 \cdot \Delta T_{in}(s) = -\alpha \cdot \left[(\omega^0 \cdot s + p) \cdot \widehat{\Delta X}(s, p) - \omega^0 \cdot \Delta X_{in}(s) \right] \quad (28)$$

Where: p – the spatial coordinate of z.

The EO hydrolysis process is carried out with full conversion of ethylene oxide in the reactor i.e. at $z=1$, $X(s, 1) \approx 0$. Considering this condition, from equation (28) it is possible to obtain the following transfer functions:

$$\frac{\Delta \bar{T}_{in}(s)}{\Delta \bar{X}_{in}(s)} = \alpha \cdot \exp(-s \cdot \tau_D) \quad (29)$$

$$\frac{\Delta \bar{T}_{out}(s)}{\Delta \bar{T}_{in}(s)} = \exp(-s \cdot \tau_D) \quad (30)$$

Where: $\tau_D = \frac{l}{\omega^0}$ - the pure time-delay, l - length of the reactor.

4. The block-flow diagram of the reactor unit

The block-flow diagram of the dynamic model of the reactor unit is represented by Fig. 2. The transfer functions W_1 , W_2 , and W_3 are determined by equation (4) while W_4 through W_{10} are determined by equations 9, 8, 19, 20, 21, 30 and 29 respectively.

5. The adequacy of the proposed dynamic model

The variables of the material balance (flow rates and levels) and temperature can be measured only in an automode conditions. Full check of adequacy of dynamic model is impractical to achieve.

Since the control actions on temperature can be realized only on the heat exchanger, the adequacy of the proposed dynamic model was checked on the heat exchanger.

The heat exchanger represents a shell and tube heat exchanger of 7.1 m length, 6 m heat exchange zone, 103 tubes with diameter of 0.02 m, 24000 kg apparatus mass. Vapor for heating is available at 3.7 bar and a flowrate of $0.046 \text{ kg} \cdot \text{s}^{-1}$. The heat exchanger to be used should be at $36 \text{ kg} \cdot \text{s}^{-1}$ of the reaction mixture.

The heat transfer coefficient is obtained from references data^[13] as $K = 1.7 \frac{\text{kJ}}{\text{m}^2 \cdot \text{s} \cdot ^\circ\text{C}}$,

The enthalpy difference between vapor and condensate at given pressure ($H_{0V} - H_{0C}$) is equal to 2145.28 kJ/kg ^[13].

Considering that the equivalent integral heat capacity of the heating agent basically is determined by the integral heat capacity of the shell as follows: $C = M_{shell} \cdot C_{shell}$, where:

$M_{shell} = 24000 \text{ kg}$ -shell mass, $C_{shell} = 0.486 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}}$ - the heat capacity of the shell^[13, 14],

Then the integral heat capacity of the shell is: $C = 11664 \text{ kJ/}^\circ\text{C}$.

Using these data it is possible to calculate the values of parameters of transfer functions 19- 21 and 23 as follows:

$$\lambda = \frac{1}{T_L} = 1.1 \text{ s}^{-1}, \tau_D = 2.93 \text{ s}, \theta = 67 \frac{\text{s} \cdot ^\circ\text{C}}{\text{kg}}, T_V^0 = 304 \text{ s}.$$

The transfer function (20) with numerical value of parameters can be represented as:

$$\frac{\Delta \bar{T}_{in}(s)}{\Delta \bar{G}(s)} = \frac{215.95 \frac{\text{s} \cdot ^\circ\text{C}}{\text{kg}}}{304 \cdot \text{s} + 1} \quad (31)$$

The vapor flowrate was increased stepwise by 50% from the initial flowrate ($0.046 \text{ kg} \cdot \text{s}^{-1}$). Thermocouples are arranged longwise of the tubular reactor from input of 0.0 m, 39.3 m, 76.5 m, and 116.7 m respectively. Fig.3 illustrates a response (F-curve) of temperature at different parts of the reactor $\Delta T_{in}(\tau)$ (at 0.0 m), $\Delta T_1(\tau)$ (at 39.3 m), $\Delta T_2(\tau)$ (at 76.5 m) and $\Delta T_3(\tau)$ (at 116.7 m). The first curve $\Delta T_{in}(\tau)$ reflecting temperature variation on an input of the hydrolysis reactor, in essence is an F-curve of the heat exchanger. The response is prolonged approximately for 5 minutes, which complies with time constant of the transfer function in equation (31). Temperature change on an output of the heat exchanger at 50% increasing vapor flowrate should make up $4.96 \text{ }^\circ\text{C}$, which is consistent with experimental results.

If the remaining thermocouples are arranged from an input of the reactor on spacing intervals of 39.3 m, 76.5 m, and 116.7 m respectively, in case of ideal plug-flow reactor, the response at given load should delay on 5.3, 10.3, 39.3 minutes respectively. The time delay has been obtained experimentally as 4.5, 6.8, and 12.2 minutes respectively. That can be explained as a result of a lengthwise stirring^[15].

It is necessary to notice that the gain in different points of the reactor is various. In the beginning of the reactor the gain is increased because of chemical reaction, and then decreased there where reaction practically is finished. This behavior is completely complying with the proposed model in equation (30). Thus, the developed dynamic models are enough adequate for synthesis of dynamic stabilization system.

6. Conclusions

An industrial process of ethylene glycols synthesis process has been considered. Linearized modelling equations of a stirred mixing tank, a heat exchanger, and a tubular reactor using a standard procedure of linearization have been developed. In order to investigate an optimal control system of this process an adequate dynamic model of ethylene oxide hydrolysis process has been proposed. All the differential equations were expressed in the transform domain form. Multivariable block-flow diagram of the dynamics process has been proposed. The adequacy of the proposed dynamic model to experimental data obtained on the industrial apparatus has been checked. It may create a system of dynamic stabilization using the proposed dynamic model.

7. References

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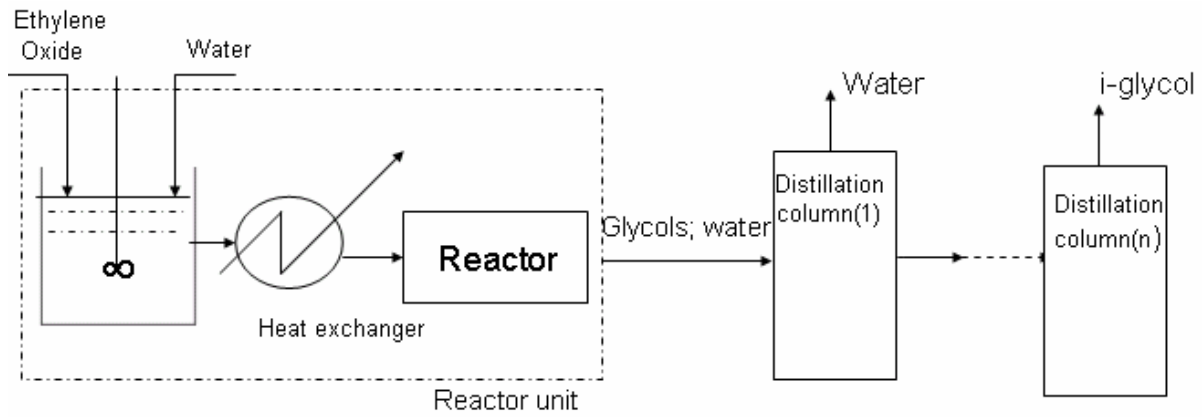


Fig. (1) A schematic flowchart of the industrial process of ethylene glycols synthesis

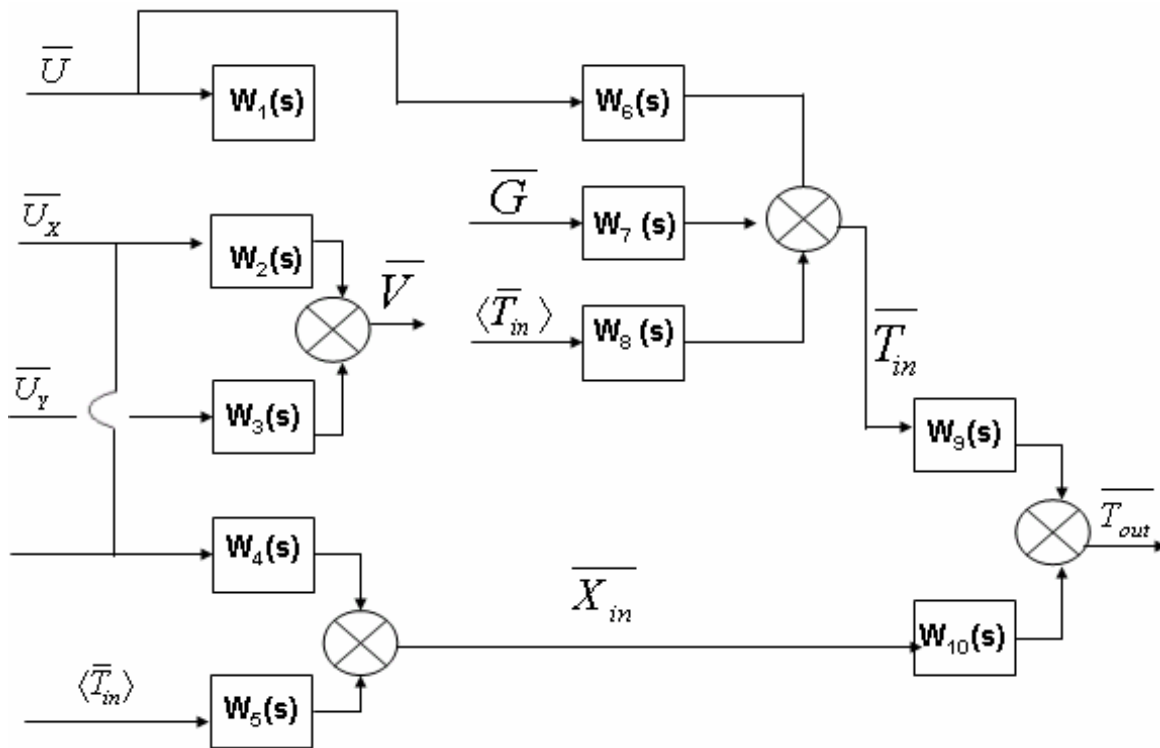


Fig. (2) The block-flow diagram of the dynamic model

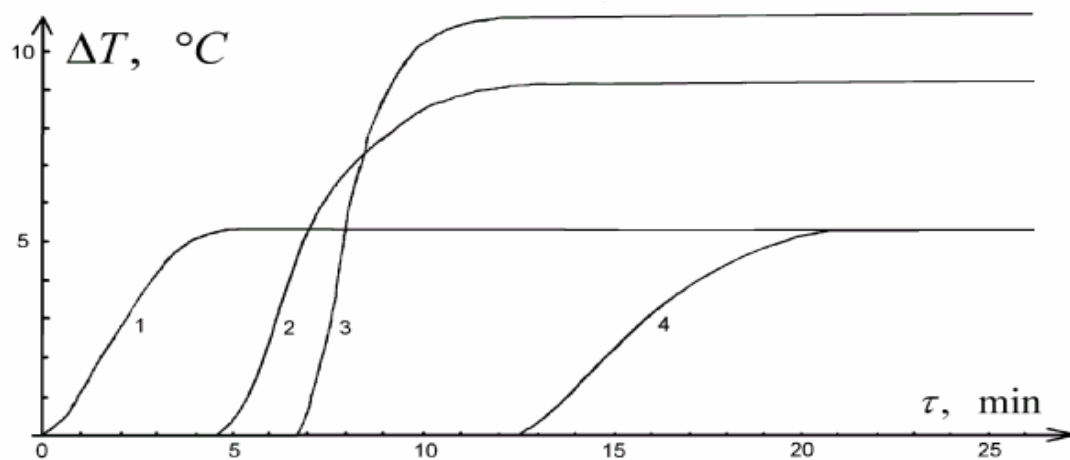


Fig. (3) Transient response of the reactor unit to a step change disturbances at distance Z of reactor's input.

1- $Z=0.0$ m 2- $Z=39.3$ m 3- $Z=76.5$ m 4- $Z=116.7$ m